

BK Restaurant System

Issue 8 – Stakeholder Needs Categorization

SENG 31242 – System Design Project

Group 01

Assigned to: Dinuja Ranaweera (SE/2022/026)

1. Introduction

This document organizes the stakeholder needs we collected during our fact-finding sessions for the BK Restaurant System project. Most of the information came from our client interview with Mr. N. C. Sanjeewa De Silva, the owner of BK Restaurant and BK Bar, who operates three branches across Ambalangoda and Kahawa in Sri Lanka.

During the interview, Mr. Sanjeewa described a range of problems he faces daily, from orders being missed through WhatsApp to having no way to track where his delivery riders are or compare how each branch is performing financially. Beyond the owner, we also identified needs from other stakeholders who will use or interact with the system: customers, delivery riders, employees (kitchen and reception), and suppliers.

Rather than leaving all these needs as a flat list, this document groups them into logical categories. This makes it much easier to map stakeholder needs to system requirements and eventually to specific modules during the design phase. Each category in this document will feed directly into the functional and non-functional requirements of the SRS.

2. Stakeholders Identified

Stakeholder	Type	Description
Restaurant Owner (Mr. Sanjeewa)	Primary	Manages all three branches. Needs oversight of orders, finances, inventory, employees, and delivery.
Customer	Primary	Places food orders online or through the app. Expects easy ordering, tracking, and promotions.
Delivery Rider	Operational	Receives delivery assignments, navigates to customer locations, updates delivery status.
Employee (Reception / Kitchen)	Supporting	Handles counter orders via POS, manages work schedule, applies for leave.
Supplier	Supporting	Receives inventory requests from the restaurant and confirms stock supply.

3. Categorized Stakeholder Needs

Category 1 – Order Management

This is the most critical category based on our interview findings. The current situation where all orders come through WhatsApp and phone calls is the biggest pain point for Mr. Sanjeewa. There is no central record, orders get missed, and there is no way to see all active orders across branches at once.

ID	Stakeholder	Need
1.1	Owner	Replace WhatsApp/phone ordering with a centralized digital system for all three branches
1.2	Owner	View all active orders across all branches in real time from a single dashboard
1.3	Customer	Browse the menu and place food orders easily through a mobile or web interface
1.4	Customer	Receive order confirmation and an estimated delivery or pickup time
1.5	Employee (Reception)	Enter walk-in customer orders quickly through a POS interface at the counter
1.6	Employee (Kitchen)	See incoming orders on a display in real time so food preparation can begin without delay
1.7	Owner	Have walk-in counter orders and online app orders merged into a single order stream

Category 2 – Delivery Management and Tracking

Mr. Sanjeewa specifically mentioned during the interview that he has no way to see where his delivery riders are or whether an order has been delivered. This category addresses all the delivery-related needs from the owner, riders, and customers.

ID	Stakeholder	Need
2.1	Owner	See the real-time GPS location of all delivery riders across branches
2.2	Owner	Receive confirmation when an order has been successfully delivered to the customer
2.3	Delivery Rider	Receive delivery assignments on a mobile app with customer address and contact details
2.4	Delivery Rider	Get turn-by-turn navigation to reach the customer location
2.5	Delivery Rider	Update the delivery status at each stage (Picked Up, On the Way, Delivered)
2.6	Customer	Track the delivery rider's location on a live map after the order is confirmed

Category 3 – Financial Management and Reporting

Currently all financial records at BK Restaurant are kept in physical books. This makes analysis almost impossible and means the owner has no way to quickly compare how each branch is performing. This category covers all the billing and reporting needs.

ID	Stakeholder	Need
3.1	Owner	Get daily, weekly, and monthly revenue reports for each branch separately
3.2	Owner	Compare revenue performance across all three branches from a single dashboard view
3.3	Owner / Employee	Automatically generate and print customer bills at the POS counter
3.4	Owner	Replace paper-based financial books with a digital financial overview
3.5	Owner	Track delivery-related revenue separately from dine-in and counter sales

Category 4 – Inventory Management

Another major problem mentioned by Mr. Sanjeewa is that inventory is managed manually and it is very easy for stock to run out without anyone realizing until it is too late. This category covers needs related to stock tracking, alerts, and supplier coordination.

ID	Stakeholder	Need
4.1	Owner	Monitor real-time stock levels for all ingredients and supplies across all branches
4.2	Owner / Employee	Receive automatic alerts when stock falls below a defined minimum level
4.3	Owner	Automatically trigger supplier order requests when stock is critically low
4.4	Employee	Manually trigger an inventory alert if a shortage is noticed during operations
4.5	Supplier	Receive inventory requests digitally and be able to confirm or update supply availability
4.6	Owner	Access an inventory history log to track usage patterns over time

Category 5 – Employee Management

BK Restaurant has staff across three branches. Currently, leave management and scheduling are handled manually. This category covers the needs of both the owner and employees for managing work-related activities through the system.

ID	Stakeholder	Need
5.1	Owner	Access a centralized employee directory covering all three branches
5.2	Owner	Review and approve or reject employee leave requests through the system
5.3	Employee	Submit leave applications online instead of through manual paperwork
5.4	Employee	View their own work schedule and upcoming shifts
5.5	Employee	Check the status of leave applications they have submitted
5.6	Owner	Assign employees to specific branches and define their roles within the system

Category 6 – Customer Engagement and Promotions

BK Restaurant currently uses its Facebook page for promotions and customer communication. The system should give the owner a more direct and controllable way to reach customers with offers and updates.

ID	Stakeholder	Need
6.1	Owner	Create and send promotional offers and discounts to customers through the app
6.2	Customer	Receive notifications about promotions, discounts, and special offers from BK Restaurant
6.3	Customer	Browse a digital menu with item descriptions, photos, and prices
6.4	Owner	Update menu items, prices, and availability at any time without technical help

Category 7 – System Access and Security

Since the system will be used by multiple types of users, the owner, employees, riders, and customers, it needs to have proper access control so each user only sees and can do what is relevant to their role.

ID	Stakeholder	Need
7.1	Owner	Role-based access control so staff can only access features relevant to their job
7.2	All Users	Secure login to prevent unauthorized access to the system
7.3	Owner	Audit log showing who performed which actions and when
7.4	Owner	Ability to add, deactivate, or modify user accounts for staff and riders

4. Needs-to-Module Mapping

The table below shows how each category of stakeholder needs maps to a planned system module. This mapping will be used in the SRS to ensure every requirement can be traced back to a real stakeholder need.

Need Category	System Module
Order Management	Online Ordering Module, POS Module
Delivery Management and Tracking	Rider Mobile App, Live Tracking Module
Financial Management and Reporting	Owner Dashboard, Billing Module
Inventory Management	Inventory Module, Supplier Portal
Employee Management	HR and Leave Management Module
Customer Engagement and Promotions	Customer App, Promotions Module
System Access and Security	Authentication and Access Control Module

5. Conclusion

Going through this categorization process helped us see clearly that the BK Restaurant System needs to cover seven distinct functional areas, all of which came directly from real problems described by Mr. Sanjeewa and the other stakeholders we identified. No single existing off-the-shelf product addresses all these needs together in a way that fits a small multi-branch Sri Lankan restaurant, which is exactly why a custom-built system is justified.

These categorized needs will now feed into the functional and non-functional requirements section of the SRS, and each category will eventually map to specific use cases, activity diagrams, and system modules during the design phase.